

Automated Recognition of Hand Gestures From Multichannel EMG Sensor Data Using Time–Frequency Domain Deep Learning for IoT Applications

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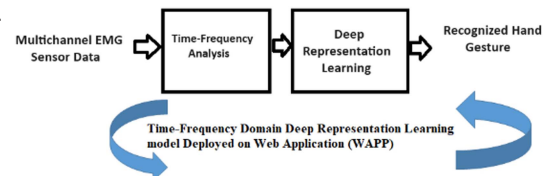
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Abstract—The automated recognition of hand gestures from multichannel electromyogram (MEMG) sensor data is essential for human–computer interaction, prosthetic control, rehabilitation, and biometric-related applications. This letter proposes a time–frequency domain deep neural network (TFDDNN)-based approach to recognize hand gestures using MEMG recordings. The MEMG recordings are segmented into frames, and the average of all channel information (mean EMG signal) for each frame is evaluated. The continuous wavelet transform is applied to the mean EMG signal to obtain the joint time–frequency representation (TFR). The TFR-based image of the mean EMG signal is used as the input to a deep representation learning network (DRLN) model to recognize hand gestures. Two publicly available databases are utilized to evaluate the performance of the proposed TFDDNN approach. The results show that the proposed TFDDNN has obtained overall accuracy values of 92.73% and 80.33% using MEMG signals from database 1 and database 2 for multiclass-based recognition of hand gestures. The suggested DRLN model has demonstrated higher accuracy for automated hand gesture recognition than existing deep-learning-based methods. We have deployed the proposed TFDDNN approach in a web application for Internet of Things (IoT)-enabled real-time recognition of hand gestures using MEMG sensor data.



Index Terms—Sensor signal processing, accuracy, continuous wavelet transform (CWT), deep representation learning, electromyogram (EMG), hand gestures, Internet of Things (IoT).

I. INTRODUCTION

The electromyogram (EMG) signal deciphers the muscle's electrical activity during contraction, and it is used for neuro-muscular disease diagnosis [1] and human–machine interface (HMI) related applications [2], [3]. The automated hand gesture recognition system using EMG sensor data is helpful in applications, such as assistive technology for elderly and disabled people, prosthetic limb control, virtual reality, biometrics, gaming, and smart home devices [4], [5]. It consists of recording multichannel EMG (MEMG) sensor data, extraction and selection of features from the MEMG data, and recognition of gestures using machine learning techniques [3]. The deep learning-based methods have the merits that these are as fully automatic as those of the manual feature-based machine learning techniques for automated gesture recognition using MEMG signals [6]. The Internet of Things (IoT)-enabled hand gesture recognition enables the real-time monitoring of EMG sensor data for HMI applications [7]. Hence, developing a new deep learning-based approach and deploying this approach on the cloud-based framework are attractive for recognizing hand gestures from MEMG data.

In recent years, various deep-learning methods have been developed to recognize hand gestures using MEMG signals. The deep convolutional neural network (CNN) model has been used for hand gesture classification using surface MEMG signals [8]. The authors in [8]

and [9] utilized the multiview deep CNN and simple attention-based models for hand gesture recognition using surface EMG data. In another work, Karnam et al. [10] suggested a hybrid deep-learning architecture by considering CNN and bidirectional long short-term memory (BI-LSTM) to classify hand gestures using MEMG signals. The deep learning model formulated using the discrete cosine transform-based attention network (DCTAN) [6] has been used for hand gesture recognition using MEMG signals. The DCTAN model has demonstrated accuracy values of 78%, which is higher than the methods reported in [6], [8], [9], and [10] in recognizing hand gestures using MEMG signals. The time-series transformer-based deep learning model has also been used to recognize hand gestures using high-density EMG signals [11].

The existing methods have considered only time-domain MEMG signals as input for recognizing hand gestures. The EMG is a non-stationary time series that varies in both time and frequency domains for different types of gesture classes [12]. Furthermore, these existing deep learning models have higher trainable parameters. Hence, the training phases of these models require higher computational resources to obtain a better representation for recognizing hand gestures using MEMG signals in the inference phase. Deep representation learning network (DRLN) has been used in various biomedical applications [13], [14]. These networks use different transfer learning models as feature extractors and a few dense layers for the classification task. The DRLN model has not been explored to recognize hand gestures using the time–frequency representation (TFR)-based analysis of MEMG signals. The novelty of this letter is developing a TFR-based DRLN framework and deploying this framework on a cloud-based web

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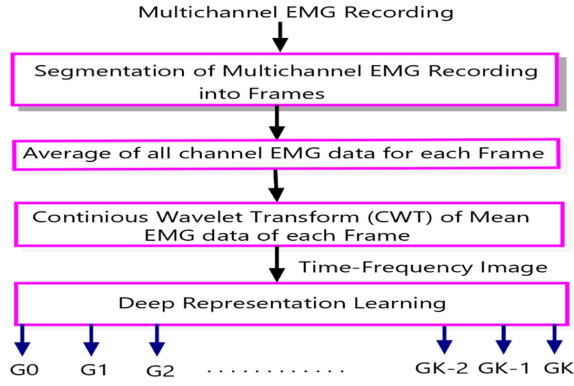


Fig. 1. Block diagram for the proposed TFDNN approach for gesture recognition using MEMG recordings.

application (WAPP) for real-time hand gesture recognition using MEMG signals.

II. MEMG SIGNAL DATABASES

In this work, we have used two publicly available databases to assess the performance of the presented TFR-based DRLN approach to recognize hand gestures. Database 1 (Mandely database) contains four-channel EMG signals (sampling frequency as 2000 Hz) from 40 subjects for ten classes (one rest class and nine hand gesture classes) of hand gestures [15]. The MEMG signals have been recorded using a four-channel BIOPAC MP36 device with the channel locations as forearm muscles for ten different types of simulating gestures performed by each subject [15]. Similarly, database 2 (Ninapro db4) contains the MEMG signals (sampling frequency as 2000 Hz) from ten subjects for 53 classes (1 rest gesture and 52 hand movement gestures) of hand gestures [16]. The gesture activities, such as finger movements, grasping movements, functional movements, and basic wrist movements, are utilized while recording MEMG signals in database 2 [16].

III. PROPOSED METHOD

The flowchart illustrating the proposed TFDNN-based approach is shown in Fig. 1. The steps involved in this flowchart are described in the following sections.

A. Frame-Based Segmentation

In this work, frame-based segmentation is utilized to divide the MEMG recordings of each subject into frames for different gesture classes. For database 1, the 6-s nonoverlapping window segments four-channel EMG recordings into MEMG signal frames. The annotations for the starting and ending samples of the MEMG recordings for each hand gesture class are given in the database [15]. Based on these points, we have evaluated the MEMG signal frames from all 40 subjects for each gesture class. For each gesture type, five MEMG signal frames are considered after the segmentation from database 1. Thus, for all 40 subjects and ten gesture classes from database 1, 2000 MEMG signal frames are evaluated. Similarly, for database 2, a sliding window of length 260 ms (520 samples) and an overlapping of 235 ms (470 samples) are utilized to segment the MEMG recording of each gesture class [6]. After segmentation, we selected only ten MEMG signal frames for each gesture class for all subjects. Therefore, for all

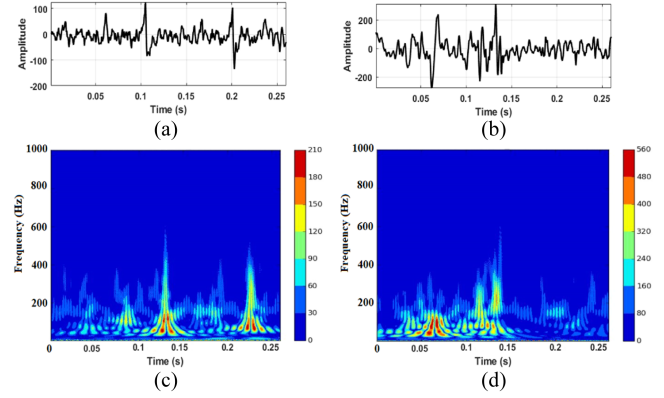


Fig. 2. (a) Mean EMG signal frame for hand gesture 6 (ring extension) class. (b) Mean EMG signal for hand gesture 43 (prismatic pinch grasp) class. (c) TFR-domain plot of Mean EMG signal frame for hand gesture 6 class. (d) TFR-domain plot of mean EMG signal frame for hand gesture 43 class.

ten subjects and 53 gesture classes, we have obtained 5300 MEMG signal frames for database 2.

B. Time-Frequency Analysis Using Continuous Wavelet Transform (CWT)

The mean or the average of all channels of the MEMG signal frame is given as $z(n) = \frac{1}{C} \sum_{c=1}^C x_c(n)$, C as total number of channels. The $x_c(n)$ is the EMG signal for c th channel. In this work, we have applied CWT over the mean MEMG signal frame. The CWT of the mean MEMG signal $z(n)$ is given as follows [17]:

$$W(s, \tau) = \frac{1}{\sqrt{s}} \sum_{\tau=0}^{N-1} z(n) \bar{\psi} \left(\frac{n - \tau}{s} \right) \quad (1)$$

where s and τ are denoted as the scale and shift, respectively. The factor $W(s, \tau)$ is denoted as the wavelet coefficients, which is the function of both scale and shift parameters. The $\bar{\psi}$ is the complex conjugate of the mother wavelet. In this work, the “Morlet”-based mother wavelet is used to evaluate the CWT of the mean MEMG signal. The CWT coefficients are a function of time (τ) and scale (s) [17]. In this letter, we have applied scale-to-frequency conversion to obtain TFR from the time-scale representation of the mean MEMG signal. The mathematical expression for scale-to-frequency conversion is given as $f = \frac{F_c}{s \times \Delta t}$ [17], where F_c is the center frequency for the Morlet-based mother wavelet function, and its value is 0.8125 Hz. The factor Δt is the sampling interval of the MEMG signal, and it is given by $\Delta t = \frac{1}{F_s}$. The total scale to evaluate the CWT of the mean MEMG signal is 1000. The mean MEMG signals for gesture 6 and gesture 43 are depicted in Fig. 2(a) and (b), respectively. Similarly, we have shown the TFR of the mean MEMG signals for gesture 6 and gesture 43 in Fig. 2(c) and (d), respectively. It is noteworthy to mention that there are significant differences in the morphologies of TFR-domain contour plots of both gesture classes. Hence, the deep learning model implemented using the input as the TFR-based images of the MEMG signals can recognize different hand gestures automatically. In this study, a new DRL model has been developed to recognize 10-classes and 53-classes-based hand gestures using TFR images extracted from the mean MEMG signals of database 1 and database 2, respectively.

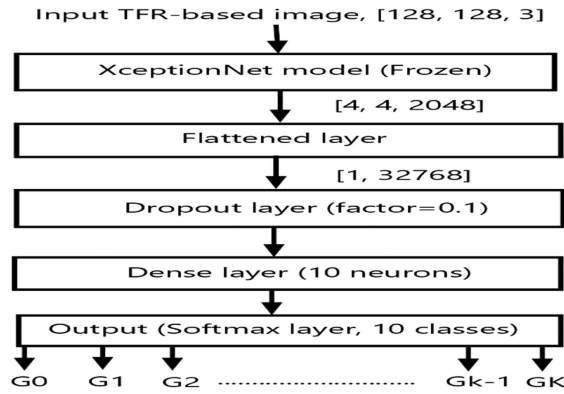


Fig. 3. Proposed DRLN model for hand gesture recognition using MEMG signals from database 1.

Table 1. DRLN Model Results With Subject-Dependent Tenfold CV

Database	Accuracy(%)	Precision(%)	Recall(%)	F1-score(%)
Database 1	95.71 $\pm$ 0.30	93.96 $\pm$ 0.56	96.52 $\pm$ 0.65	95.29 $\pm$ 0.22
Database 2	86.56 $\pm$ 0.24	85.16 $\pm$ 0.45	85.21 $\pm$ 3.81	86.23 $\pm$ 0.18

C. Deep Representation Learning Network

The architecture of the DRLN model is depicted in Fig. 3. The input to this model is the TFR-based image with size as $128 \times 128 \times 3$. The DRLN model uses a frozen XceptionNet-based transfer learning block [18] to obtain the representation or feature map from the TFR-based image of the mean MEMG signal frame. The size of the tensor obtained after the frozen XceptionNet is given as $4 \times 4 \times 2048$ [18]. After the frozen XceptionNet block, the flattened layer obtains a feature vector of size 1×32786 . The dropout layer, dense layer, and softmax layer (output) are used after the flattened layer in the suggested DRLN model for the automated recognition of hand gestures. The number of trainable and nontrainable parameters for the DRLN model are 327 800 and 20 861 480, respectively. For the database 1 case, the dense layer after the dropout layer in the DRLN model architecture contains ten neurons. However, for the database 2 case, as we have 53 classes, the number of neurons in the dense layer is 512. The hyperparameters used during the training of the DRLN model are the number of epochs as 13, batch size as 32, optimizer as “Adam,” and loss function as “sparse categorical cross-entropy,” respectively [13]. The number of neurons in the dense layer and the hyperparameters of the DRLN model are selected using the grid-search-based approach. In this study, the training and test MEMG signal frames are selected based on both subject-dependent (hold-out validation and fivefold cross-validation (CV) [19] and subject-independent validation (SIV) techniques [6]. For SIV cases, 70% and 30% of the MEMG signal frames from different subjects are used to train and test the proposed DRLN model for recognizing hand gestures. The MEMG frames from subject numbers 3, 18, 21, 24, 31, and 36 of database 1 and subject numbers 2, 5, and 8 from database 2 are used in the testing phase of the DRLN model for the SIV case.

IV. RESULTS AND DISCUSSION

The classification metrics of the proposed TFDNN-based approach are given in Table 1 using TFR-based images of MEMG signals from database 1 and database 2 cases with the subject-dependent fivefold CV method. It has been observed that the average accuracy values of the DRLN model for database1 and database2 MEMG signal cases are 95.71% and 86.56%, respectively. Similarly, for the subject-dependent

Table 2. DRLN Model Results With SIV

Database	Accuracy(%)	Precision(%)	Recall(%)	F1-score(%)
Database 1	92.73	89.34	91.62	87.94
Database 2	80.33	78.84	81.67	79.82

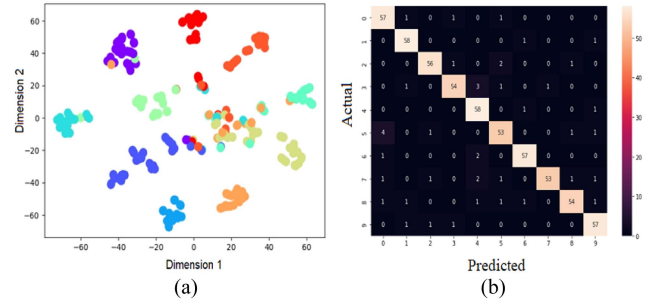


Fig. 4. (a) t-SNE plot computed using the feature vectors of the dense layer (before output layer) of DRLN model for test MEMG frames from database 1. (b) Confusion matrix of DRLN model obtained using test MEMG frames from database 1.

hold-out validation case, the accuracy values of the DRLN models are obtained as 97.43% and 84.93% using TFR-based images of MEMG signals from database 1 and database 2, respectively. We have computed the classification results of the suggested DRLN model using the SIV-based selection of TFR-based images of MEMG signals during the training and testing phases. These new results are depicted in Table 2. For the SIV case, the DRLN model has produced accuracy values of 92.73% and 80.33%, respectively. The DRLN model has demonstrated higher classification performance in all measures using MEMG signals from database 1 compared with database 2 cases for subject dependent validation (SDV) and SIV techniques. Database 2 has more hand gesture classes (53) than ten hand gesture classes in database 1. The MEMG signals are highly correlated for gesture classes, such as wrist supination and wrist pronation. This correlated information has been captured in the TFR-based images of the gesture classes. Hence, the proposed DRLN model has produced a lower accuracy value using TFR-based images from MEMG signals of database 2.

We have shown the t-SNE plot of the feature vectors evaluated at the dense layer before the output layer of the proposed DRLN model for test MEMG instances in Fig. 4(a) to recognize hand gestures using MEMG signals from database 1. It is seen that the learned features in the dense layer of the proposed DRLN model are found to be discriminative for the different gesture classes. The confusion matrix of the DRLN model using the test MEMG signals from database 1 is shown in Fig. 4(b). It is observed that the number of true positives of the DRLN model is 557 out of 600 MEMG test instances from database 1 using the SIV strategy. The XceptionNet uses depthwise and pointwise separable convolutions compared with regular convolution blocks [18]. We have considered other frozen transfer learning blocks, such as VGG16, ResNet50, DenseNet201, and EfficientNetV2B2 [20], instead of the XceptionNet block in the proposed DRLN model to recognize hand gestures. The overall accuracy values of DRLN with VGG16, DRLN with Resnet50, DRLN with DenseNet201, and DRLN with EfficientNetV2B2 are obtained as 28.56%, 31.44%, 34.36%, and 42.72%, respectively, which are less than DRLN with XceptionNet in recognizing hand gestures using TFR-images of mean MEMG signals from database 1. We have also calculated the results of the proposed DRLN model with a fine-tuned XceptionNet block to recognize hand gestures. The accuracy values of the fine-tuned XceptionNet-based DRLN models are obtained as 42.31% and 33.87% using TFR-images

Table 3. Comparison With Existing Deep Learning-Based Methods to Recognize Hand Gestures Using MEMG Signals From Database 2

Method considered	Accuracy (%)
Multi-view CNN model [8]	58.00
Hybrid model (CNN and Bi-LSTM) [10]	78.92
Attention-based model [9]	76.23
DCTAN model [6]	78.70
Proposed work (TFD-based DRLN)	80.33



Fig. 5. TFDNN approach deployed on WAPP for real-time recognition of hand gestures using MEMG signals.

from mean MEMG signals from database 1 and database 2, respectively, for the SIV case. The drop in the accuracy value of fine-tuned XceptionNet-based DRLN is due to the overfitting issue in the training of such an extensive network (all layers of XceptionNet).

The proposed approach has been compared with existing deep learning-based methods for hand gesture recognition using MEMG signals from database 2 in Table 3. It has been noted that the proposed TFD-based DRLN approach has demonstrated higher accuracy value than various deep learning models to recognize hand gestures with SIV strategy [6], [8], [9], [10]. We have also evaluated discrete wavelet transform (DWT)-domain [12] energy features of MEMG signals from database 2 and used the support vector machine classifier [4] to recognize hand gestures. The accuracy value of the DWT-based method is obtained as 72.31%, which is less than the proposed time-frequency domain deep neural network (TFDDNN) approach. The suggested DRLN model has fewer training parameters than the existing deep learning-based networks for multiclass-based recognition of hand gestures.

The accuracy of the DRLN model is obtained as highest using CWT-based TFR images of mean MEMG signals with “Morlet” mother wavelet as compared with “Mexican-hat” and “Shannon” cases. For this reason, the “Morlet”-based mother wavelet is chosen for CWT-based analysis of MEMG signals. We have quantized the DRLN model with floating point 16-bit (FP16) and integer 8-bit (INT8) cases to recognize hand gestures. The accuracy values of DRLN models with FP16 and INT8 quantization cases using MEMG signals from database 1 are 87.74% and 82.32%, respectively. Hence, we have deployed the floating point 32-bit version of the proposed TFDDNN approach on a WAPP for IoT-enabled real-time recognition of hand gestures. The hand gesture prediction for an MEMG signal on WAPP is shown in Fig. 5. The advantages of the proposed work include real-time monitoring, accessibility, and scalability. Similarly, the high latency, privacy, and security of MEMG data from the subjects and dependency on internet connectivity are the limitations of the proposed method for IoT-enabled hand gesture recognition.

V. CONCLUSION

A time-frequency domain deep learning-based approach has been proposed in this work to recognize hand gestures using MEMG signals. The TFR-based image has been evaluated using channel averaging of

the MEMG signal and CWT. The DRLN model has been employed to recognize hand gestures automatically using TFR images of MEMG signals. The proposed approach has produced accuracy values of 95.71% and 86.56% for the 10-classes and 53-classes-based SDV for hand gesture recognition using MEMG signals. For the SIV-based hand gesture recognition task, the suggested approach has demonstrated accuracy values of 92.73% and 80.33% for 10-classes and 53-classes-based recognition of hand gestures.

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